

Hamza Idrees

BACKEND DEVELOPER

hamzaidrees083@gmail.com | +92 306 7566528 | linkedin.com/in/hamzaidrees2002 | hamza-idrees.netlify.app

SUMMARY

Backend Developer with **1.5+ years** of production experience building multi-tenant SaaS APIs serving **400+ businesses** across **1,000+ locations** in **35+ countries**. Specialized in scalable background-job systems and AI / RAG integrations (OpenAI, Anthropic Claude). Strong across the full backend lifecycle — database design, API development, AWS infrastructure, and automated testing at **90%+ coverage**.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python
Backend	Node.js, Express.js, FastAPI, BullMQ, node-cron, REST, Swagger/OpenAPI
Databases	PostgreSQL (Sequelize), MongoDB (Mongoose), Redis, Qdrant (vector search)
Cloud & AI	AWS Lambda, S3, SQS, SES · OpenAI, Anthropic Claude, RAG pipelines, embeddings
Auth & Integrations	JWT, OAuth 2.0, Microsoft Entra ID (SSO), DeepL, SendGrid, Google APIs
Testing	Jest, Mocha, Supertest
Frontend	React, Next.js, TypeScript, Tailwind CSS, Redux Toolkit

PROFESSIONAL EXPERIENCE

Backend Developer · Global Software Consulting Dec 2024 – Present

Obenan — SaaS for business location management & online reputation across Google, Facebook & 20+ review platforms

- Built and maintained **multi-tenant REST APIs** (Node.js / Express) over a dual-database architecture — **PostgreSQL (Sequelize, 100+ models)** and MongoDB — adding Redis caching to high-traffic endpoints to eliminate redundant queries.
- Engineered an **AWS Lambda** function to fetch **40,000+ reviews** within the 15-minute execution limit via optimized batching and pagination.
- Developed AI sentiment-analysis & review-response pipelines (OpenAI, Anthropic Claude) and a batched post-generation pipeline processing **100+ posts in <30s**, cutting generation time by **90%+**.
- Built and maintained **10+ BullMQ / Redis job queues** for async review ingestion, analysis, report exports, and email delivery; automated two-way Google / Facebook listing sync.
- Implemented **JWT auth + RBAC**, OAuth 2.0 (Google, Facebook) and led **Microsoft Entra ID (Azure AD) SSO**; enforced rate limiting, input validation, and webhook signature verification.
- Wrote unit & integration tests (Jest, Mocha, Supertest) contributing to **90%+ code coverage**.

Stack: Node.js, Express, PostgreSQL, MongoDB, Redis, BullMQ, AWS Lambda/S3/SQS/SES, JWT, OAuth 2.0, React, TypeScript

Lumen — production RAG system letting clinic staff query ~5,000 patient records in plain English with cited answers

- Designed an end-to-end **RAG pipeline** (Python / FastAPI): GPT-4o-mini intent extraction, ada-002 embeddings, and Qdrant vector store — **sub-100ms vector search** and **<2s first-token** streaming over SSE.
- Built a **20+ filter query parser** (condition, age, doctor, dates, billing, blood type) that cut irrelevant retrievals by **60%+** vs. unfiltered semantic search.
- Added cosine-similarity thresholding and a "don't know" fallback, **reducing hallucination to near-zero** on out-of-scope queries; supported bulk ingestion of 5,000 records in <60s.

Stack: Python, FastAPI, Qdrant, OpenAI (GPT-4o-mini, ada-002, GPT-4.1-mini), SSE, Next.js 14, Tailwind CSS, Framer Motion

PROJECTS

ShipEase — Logistics Management Platform React · Node.js · PostgreSQL · MUI

- Full-stack logistics platform connecting service providers, drivers, and users with real-time shipment tracking; built the **web app, backend API, and mobile application** for both user and driver roles.

EDUCATION

B.S. in Software Engineering — FAST — National University of Computer & Emerging Sciences 2020 – 2024